WEEK 1 · LESSON 3 OF 24 · HANDOUT

Generative AI and Large Language Models

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=LPZh9BOjkQs — open the lesson online to watch inside the course.

🎓 **Go deeper:** Andrej Karpathy: Intro to Large Language Models (1-hour talk) — https://www.youtube.com/watch?v=zjkBMFhNj_g

Your goal in this lesson: Explain what generative AI is, describe in simple terms how a large language model produces text, and list what these tools can create.

From recognising to creating

Earlier AI mostly **recognised** or **predicted**: "this email is spam", "this photo shows a cat". **Generative AI** goes further — it **creates new content**: text, images, audio, video and computer code.

When you ask ChatGPT to write a cover letter, or ask an image tool to draw "a modern Kigali office at sunset", you are using generative AI.

How a large language model writes

A **large language model (LLM)** is trained on an enormous amount of text — books, websites, articles, code. During training it plays one simple game billions of times: **guess the next word**.

> *"The capital of Rwanda is ___" → most likely next word: "Kigali".*

By getting extremely good at predicting the next word (more exactly, the next small piece of text called a **token**), the model learns grammar, facts, writing styles and even some reasoning patterns. When you ask a question, it writes the answer one token at a time, each time choosing a likely next token.

Two important consequences:

1. **It is very fluent.** It has seen so much language that its writing sounds natural and confident.
2. **Fluent is not the same as true.** The model picks *likely* words, not *verified* facts. That is why hallucinations happen.

Key ideas in plain words

- **Prompt** — the instruction or question you give the AI.
- **Token** — a small piece of text (a word or part of a word) that the model reads and writes.
- **Context window** — how much text the model can "keep in mind" at once in one conversation.
- **Training data cut-off** — the date after which the model has no built-in knowledge, unless the tool can search the web.
- **Multimodal** — a model that can work with more than text, such as reading a photo of a receipt or listening to audio.

What generative AI can create

Type	Example tools	Example use
Text	ChatGPT, Gemini, Claude, Copilot	Emails, reports, summaries, lesson plans
Images	Image generators in ChatGPT, Gemini, Canva	Posters, social-media graphics
Audio	Voice and music tools	Voice-overs, podcast clean-up
Video	Video generators and editors	Short promotional clips
Code	GitHub Copilot, AI assistants	Formulas, scripts, websites

Where it shines — and where to be careful

Shines at: first drafts, summarising long text, explaining ideas at different levels, brainstorming, translating, re-formatting.

Be careful with: exact numbers and dates, recent news, quotations, legal or medical advice, and anything about Rwanda that is very local or very new. Always check.

Remember this: Generative AI creates new content by predicting likely next pieces; that makes it fluent and fast — but not automatically correct.

Name: _____ Date: _____

Activity

Task 1

Ask an AI assistant: *"Write a 100-word welcome message for new students joining a computer course in Kigali."*

Task 2

Then ask it to rewrite the same message **(a)** for parents, **(b)** in a more formal tone, and **(c)** as three short bullet points.

Task 3

Note how the same model changes style when you change the instruction. Which version is best, and why?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What makes generative AI different from earlier AI?
   A. It works without electricity
   B. It cannot use language
   C. It only recognises images
   D. It creates new content such as text, images and code
2. How does a large language model produce an answer?
   A. It searches a single correct answer in a database
   B. It writes one token at a time, choosing likely next tokens
   C. It copies a website word for word
   D. A person types the answer in real time

3. Why can a fluent AI answer still be wrong?

   A. It cannot be wrong if it is fluent
   B. Because the user typed in lowercase
   C. Because the model chooses likely words, not verified facts
   D. Because the internet was slow

4. What is a 'prompt'?

   A. The instruction or question you give the AI
   B. The answer the AI gives
   C. The AI's training data
   D. A type of computer chip

5. A model that can read a photo of a receipt and answer questions about it is…

   A. Unsupervised
   B. Multimodal
   C. Offline
   D. Rule-based

6. For which task should you be MOST careful and check the AI's output?

   A. Rewriting a paragraph more simply
   B. Exact figures from a recent government report
   C. Changing the tone of an email
   D. Brainstorming poster ideas

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